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BRIDGING THE FUNDING GAP: STRENGTHENING AGRICULTURAL R&D FOR FOOD SECURITY IN NIGERIA

Introduction

Research and Development (R&D) investment is widely recognized as a key driver of technological advancement (Tsai et al., 2009). In agriculture, domestic R&D plays a crucial role in developing innovative solutions such as drought-resistant crops, efficient farming techniques, and post-harvest loss reduction strategies, all of which contribute to sustainable agricultural growth. Despite its importance, global agricultural R&D faces significant funding gaps, particularly in developing nations (Rosegrant et al., 2022). This raises critical questions: How are R&D projects in Nigeria's public agricultural research institutes funded? Who are the major stakeholders? Is the current funding model effective, and what improvements are necessary for better implementation? This policy brief examines the funding landscape for agricultural R&D in Nigeria's federal research institutions and proposes actionable strategies to bridge the existing funding gap.

Method

Using a systematic approach, this 2024 study identified 50 crop-based R&D projects undertaken by federal agricultural research institutes in Nigeria, as recorded in the National Annual Budgetary Reports from 2018 to 2022. To evaluate their funding, a well-structured questionnaire was administered to the principal investigators of these projects. The findings were analyzed and discussed in this brief.

Key Findings

A major challenge identified in the study is the inadequate release of allocated funds for R&D projects. Findings revealed that the majority of these projects received less than 50% of their approved budget, severely limiting their implementation and impact. This funding shortfall could hinder progress in agricultural innovation, delaying critical research efforts aimed at improving productivity and food security.

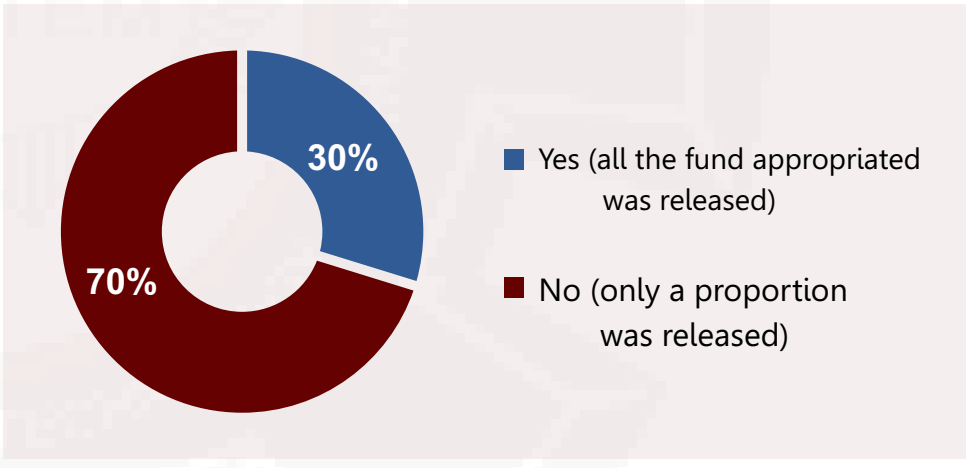


Fig. 1: Percentage share of project fund released during project implementation

Another key finding is the prevalence of underfunded R&D projects, with approximately 75% operating on insufficient budgets (Fig. 2). This funding gap often results in incomplete research, stalled innovations, and limited

scalability of successful pilot projects. For example, a post-harvest loss reduction initiative may show promising outcomes but fail to progress to large-scale implementation due to financial constraints. This cycle of underachievement prevents R&D breakthroughs from translating into tangible benefits for farmers and the agricultural sector, ultimately hindering Nigeria's progress toward achieving the Sustainable Development Goal (SDG) of zero hunger.

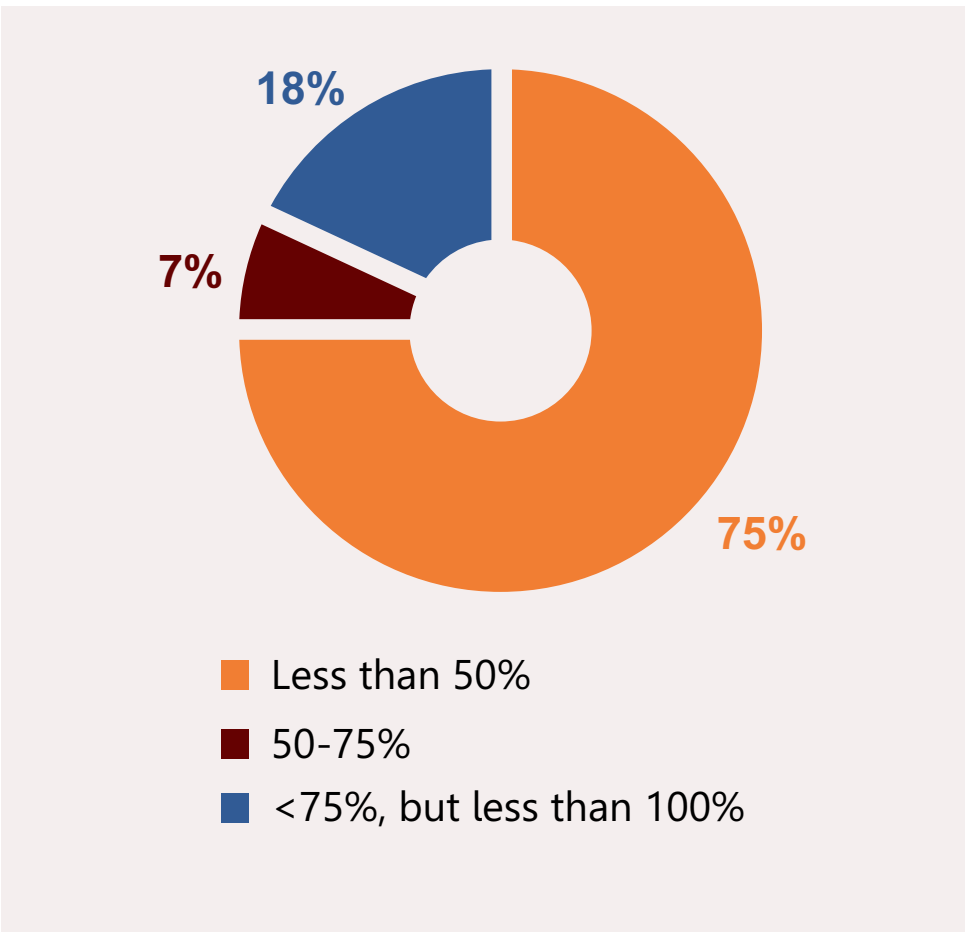


Fig. 2: Percentage share of partially funded R&D project

Only 3% of the R&D projects received external funding support, notably from non-governmental organizations (NGOs) or civil society organizations (CSOs). This indicates an overwhelming reliance on government funding, which has proven to be inconsistent and insufficient for sustaining long-term research and innovation. The lack of diverse funding sources limits the scope and impact of R&D initiatives, reducing opportunities for collaboration, technology transfer, and capacity-building.

Without active private sector and donor involvement, agricultural research efforts in Nigeria may continue to struggle with delays, incomplete studies, and inadequate scaling of innovations. Expanding funding partnerships is essential to ensuring sustainable agricultural advancements and achieving national food security goals.

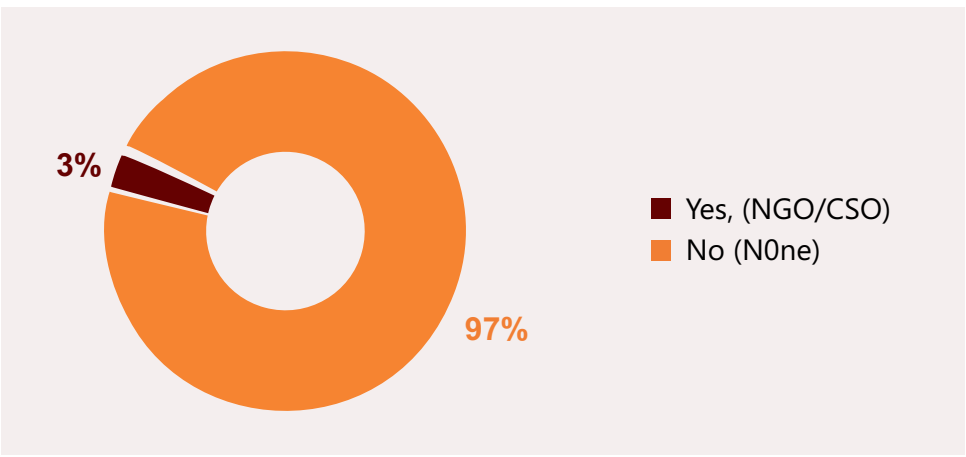
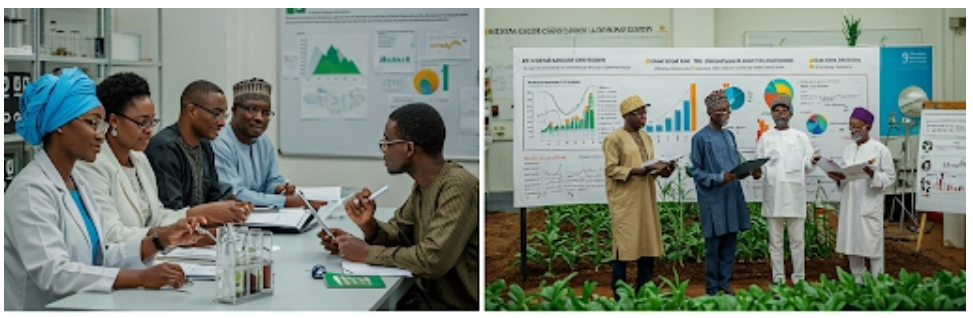


Fig. 3: Share of R&D projects that received external Funding

Conclusion

Enhancing the funding of agricultural R&D projects is crucial for strengthening food security in Nigeria. This brief highlights the importance of timely and sufficient fund allocation to research institutions. To bridge the funding gap, it is essential to diversify funding sources, adopt performance-based financing, and uphold financial accountability. Collaboration among policymakers,



stakeholders, and development partners is key to establishing a sustainable financing framework that supports agricultural R&D. By addressing these challenges, Nigeria can empower researchers in the agriculture ecosystem; boost agricultural productivity; and ensure a more food secured future.

Policy Recommendation

To strengthen agricultural R&D funding and enhance food security, the following policy actions are recommended:

i. Establish a legal framework mandating the full and timely disbursement of appropriated funds.
ii. Improve budgetary processes to ensure research institutions receive funds as scheduled.
iii. Encourage private sector investment in agricultural R&D through tax incentives and public-private partnerships.
iv. Strengthen collaborations with international organizations while ensuring alignment with national agricultural goals.
v. Promote grant-writing and fundraising capacity among research institutions.
vi. Develop monitoring and evaluation mechanisms to track research progress and financial utilization.

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